End-Term Examination

(CBCS) (SUBJECTIVE TYPE)

B. Tech (CSE/IT/CSE AI), 4th Semester

(May, 2022)

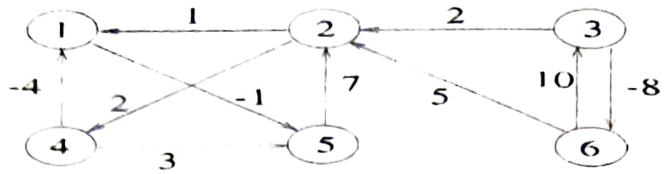
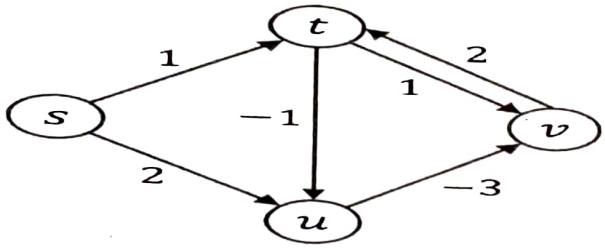
Q.P Code 220517

Subject Code: BCS 204	Subject: Design and Analysis of Algorithms

Time : 3 Hours **Maximum Marks : 60**

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.		**(5*4=20)**
	(a) Suppose we are given two n-element sorted arrays A and B that should not be viewed as sets (that is, A and B may contain duplicate entries). Describe an O(n)-time method for computing an array representing the set A ∪ B (with no duplicates).	
	(b)Design a branch and bound algorithm for solving 0/1 Knapsack problem. Explain with an example.	
	(c) Show that the depth first search traversal of a graph can be used to find the strongly connected component of a graph. Explain with an example also	
	(d) You are given an array A[1..n] that is first decreasing and then increasing. More precisely, there is a coordinate $1 \leq p \leq n$ such that for all $i < p$, $A[i] > A[i + 1]$, and for all $i \geq p$, $A[i] < A[i + 1]$. Your goal is to find the smallest element in this array. Design O(log n) time algorithm.	
	UNIT-1	
Q2.	An investment company has stock prices for n consecutive days. That is for each day, they have a specific stock price. They want to know on which day they should buy a stock and sell it on some other day so that they make the maximum profit out of it. Hint: Think of it as an array A[0..n] of distinct integers, you need to find two indices i and j such that (j > i) and A[j] − A[i] is maximum. Design a O(n log n) algorithm for this problem.	**(10)**
Q3.	Suppose you are choosing between the following 3 algorithms: • Algorithm A solves the problem of size n by dividing it into 5 subproblems of size n/2, recursively solving each subproblem, and then combining the solutions in linear time. • Algorithm B solves the problem of size n by recursively solving two subproblems of size n − 1 and then combining the solutions in constant time • Algorithm C solves the problem of size n by dividing it into nine subproblems of size n/3, recursively solving each subproblem, and then combining the solutions in O(n2) time. What are the running times of each algorithm and which would you choose and why?	**(10)**
	UNIT-2	

Q4.	Run the Floyd Warshall algorithm on the weighted directed graph shown in the figure. Show the matrix D(k) that results after each iteration 	**(10)**
Q5.	a. We are given a set of n characters along with their frequencies. We are required to represent each character by a unique code word using 0s and 1s such that no code word is a prefix of another. the goal is to find such code words that achieve maximum compression . Write a greedy algorithm and analyse its complexity b. Construct an optimal binary code for four characters a,b, c, d with frequencies 14,3,6,10 respectively. How many bits are needed to encode a string containing 14 a's, 3 b's, 6 c's and 10 d's using this code.? How many bits would be needed if we used 2 bits for each character? Which code is better	**(5+5=10)**
	UNIT-3	
Q6.	(a)Run the Bellman-Ford algorithm on the following graph to find the shortest path from s to every other node. Show the table that you fill in while running the algorithm.  (b) Now suppose you ran one extra iteration of Bellman-Ford. How could you use that last iteration to detect whether the graph has a negative-weight cycle?	**(8+2=10)**
Q7.	Suppose that we are given a cable network of n sites connected by communication channels. Unfortunately, the communication channels are not perfect and they may fail with certain probability . Let f(u,v) denote the failure probability for the channel between sites u and v. One of the n sites is the central station and your problem is to find the most reliable paths (paths with lowest failure probability) from the central station to all other sites. Design a greedy algorithm and analyse its complexity.	**(10)**
	UNIT-4	
Q8.	a. Explain the string matching algorithm using finite state automata. b. Search the pattern 'AGCTA' in the string 'AAGCCTTAGCTAATG' using this algorithm.	**(5+5=10)**
Q9.	a. Write a short note on NP- Complete problems b. Prove that the clique problem is polynomial-time reducible to the vertex cover problem.	**(4+6=10)**